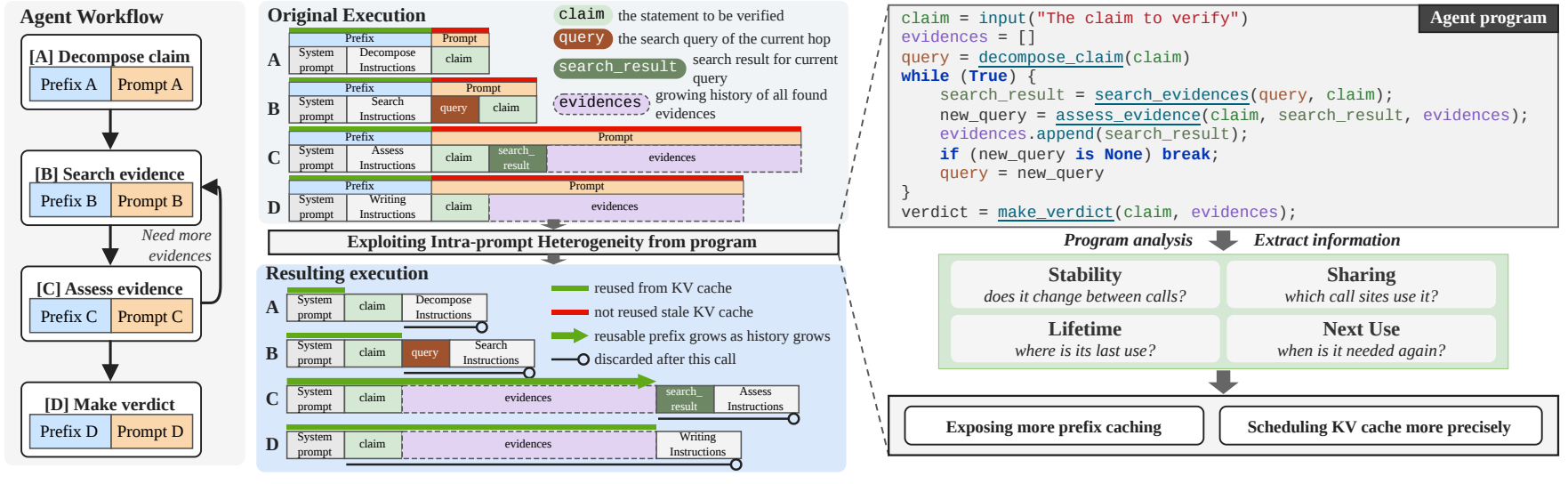

Agent Workflow
[A] Decompose claim
Prefix A
Prompt A
[B] Search evidence
Prefix B
Prompt B
Need more evidences
[C] Assess evidence
Prefix C
Prompt C
[D] Make verdict
Prefix D
Prompt D
Original Execution
claim the statement to be verified
query the search query of the current hop
search_result search result for current query
evidences growing history of all found evidences
A
B
C
D
System prompt
Decompose Instructions
Search Instructions
Assess Instructions
Writing Instructions
Prefix
Prompt
claim
query
search_result
evidences
Exploiting Intra-prompt Heterogeneity from program
Resulting execution
reused from KV cache
not reused stale KV cache
reusable prefix grows as history grows
discarded after this call
Agent program
claim = input("The claim to verify")
evidences = []
query = decompose_claim(claim)
while (True) {
search_result = search_evidences(query, claim);
new_query = assess_evidence(claim, search_result, evidences);
evidences.append(search_result);
if (new_query is None) break;
query = new_query
}
verdict = make_verdict(claim, evidences);
Program analysis
Extract information
Stability
does it change between calls?
Sharing
which call sites use it?
Lifetime
where is its last use?
Next Use
when is it needed again?
Exposing more prefix caching
Scheduling KV cache more precisely